Chapter II-1

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1.2. (a) Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X, and $p \in X$ a point. Show that $(\ker \varphi)_p \cong \ker(\varphi_p)$ and $(\operatorname{im} \varphi)_p \cong \operatorname{im}(\varphi_p)$.
(b) Show that φ is injective (resp. surjective) if and only if φ_p is injective (resp. surjective) for all $p \in X$.
(c) Show that a sequence $\cdots \to \mathcal{F}_{i-1} \xrightarrow{\varphi_{i-1}} \mathcal{F}_i \xrightarrow{\varphi_i} \mathcal{F}_{i+1} \to \cdots$ of sheaves is exact if and only if, for all $p \in X$, the associated sequence of stalks is exact.

Proof. (a) A germ $(s, U) \in (\ker \varphi)_p \subset \mathcal{F}_p$ iff there exists $p \in V \subset U$ with $\varphi(U)(s)|_V = 0$. That is,

$$\varphi_p(s, U) = (\varphi(V)(s)|_V, V) = (0, V).$$

Similarly, a germ $(s, U) \in (\operatorname{im} \varphi)_P \subset \mathcal{G}_p$ iff there exists $p \in V \subset U$ with $s|_V = \varphi(V)(t)$. That is,

$$(s, U) = (s|_V, V) = (\varphi(V)(t), V) = \varphi_p(t, V).$$

(b) φ is injective iff $\ker \varphi \cong 0$. The zero map $\ker \varphi \to 0$ is an isomorphism iff each of its localizations $(\ker \varphi)_p \to 0$ are isomorphisms. Using (a), we see that φ is injective iff $\ker(\varphi_p) \cong (\ker \varphi)_p \cong 0$, i.e. φ_p injective, for all $p \in X$.

φ is surjective iff $\mathcal{G} = \operatorname{im} \varphi$. That is, the inclusion map $i : \operatorname{im} \varphi \to \mathcal{G}$ is an isomorphism. This occurs iff each $(\operatorname{im} \varphi)_p \to \mathcal{G}_p$ is an isomorphism, which occurs iff each $\operatorname{im}(\varphi_p) \xrightarrow{\sim} (\operatorname{im} \varphi)_p \to \mathcal{G}_p$ is an isomorphism. This map is the inclusion $\operatorname{im}(\varphi_p) \to \mathcal{G}_p$, so the result follows.

(c) It suffices to show the result for short exact sequences $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$. (b) gives us φ injective iff each φ_p is injective, ψ surjective iff each ψ_p is surjective.

If $\operatorname{im} \varphi = \ker \psi$, then this induces natural isomorphisms $\operatorname{im} \varphi_p \cong \ker \psi_p$ for all $p \in X$. Inclusions of sheaves induce inclusions of stalks, so $\operatorname{im} \varphi_p = \ker \psi_p$.

Suppose instead that $\operatorname{im} \varphi_p = \ker \psi_p$ for all $p \in X$, and fix $U \subset X$, $s \in \mathcal{F}'(U)$. For each $p \in U$, we have $\widetilde{\operatorname{im}} \varphi_p = \ker \psi_p$, so $\psi_p \varphi_p(s, U) = (0, U)$. That is, there is $p \subset V_p \subset U$ such that

$$\psi(U)\varphi(U)(s)|_{V_p} = 0.$$

Then the V_p form a cover of U, so the sheaf axiom gives us $\psi(U)\varphi(U)(s) = 0$, whence

$$\widetilde{\operatorname{im}} \varphi \subset \ker \psi.$$

Then, in fact, there is an inclusion $i : \operatorname{im} \varphi \hookrightarrow \ker \psi$ which induces the inclusions $i_p : \operatorname{im} \varphi_p \hookrightarrow \ker \psi_p$ for each $p \in V$. The latter inclusions are isomorphisms, so it follows that the former is an isomorphism as well. That is,

$$\operatorname{im} \varphi = \ker \psi.$$

$\square$

1.5. Show that a sheaf morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is an isomorphism iff it is injective and surjective.

Proof. φ is an isomorphism iff each $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is an isomorphism. This occurs iff each φ_p is injective and surjective, which occurs iff φ is injective and surjective by the 2b. $\square$

1.8. Show that, for any $U \subset X$ open, the functor $\Gamma(U, -) : \mathrm{Sh}(X) \to \mathfrak{Ab}$ is left-exact.

Proof. Suppose $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ is exact. Then $\ker(\varphi(U)) = (\ker\varphi)(U) = 0$ gives us that $\varphi(U)$ is injective. It suffices to show that $\operatorname{im}(\varphi(U)) = \ker(\psi(U))$. Since $\ker\psi = \operatorname{im}\varphi = (\widetilde{\operatorname{im}}\,\varphi)^+$, it suffices to show that $\widetilde{\operatorname{im}}\,\varphi$ is a sheaf. $\varphi(U)$ is injective, so $\varphi(U) : \mathcal{F}'(U) \to \widetilde{\operatorname{im}}\,\varphi(U)$ is an isomorphism of abelian groups. Then $\varphi : \mathcal{F}' \to \widetilde{\operatorname{im}}\,\varphi$ is an isomorphism of presheaves, so $\widetilde{\operatorname{im}}\,\varphi$ is a sheaf. $\square$

1.16. *Flasque Sheaves.* A sheaf $\mathcal{F}$ on a topological space X is *flasque* if for every inclusion $V \subset U$ of open sets, the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$ is surjective.

(a) Show that a constant sheaf on an irreducible topological space is flasque.

(b) If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ is flasque, then for any open set U, the sequence $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U) \to 0$ of abelian groups is also exact.

(c) If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ and $\mathcal{F}$ are flasque, then $\mathcal{F}''$ is flasque.

(d) If $f : X \to Y$ is a continuous map and $\mathcal{F}$ is a flasque sheaf on X, then $f_*\mathcal{F}$ is a flasque sheaf on Y.

(e) Let $\mathcal{F}$ be any sheaf on X. We define a new sheaf $\mathcal{G}$, called the sheaf of *discontinuous sections* of $\mathcal{F}$ as follows. For each open set $U \subset X$, $\mathcal{G}(U)$ is the set of maps $s : U \to \bigcup_{p\in U} \mathcal{F}_p$ such that for each $p \in U$, $s(p) \in \mathcal{F}_p$. Show that $\mathcal{G}$ is a flasque sheaf, and that there is a natural injective morphism $\mathcal{F} \hookrightarrow \mathcal{G}$.

Proof. $\square$

1.19. *Extending a Sheaf by Zero.* Let X be a topological space, Z a closed subset, $i : Z \to X$ the inclusion, $U = X - Z$ the complimentary open subset, and $j : U \to X$ its inclusion.

(a) Let $\mathcal{F}$ be a sheaf on Z. Show that $(i_*\mathcal{F})_p$ is $\mathcal{F}_p$ if $p \in Z$, and 0 if $p \notin Z$.

Proof. $\square$

1.20.